

He who refuses to do arithmetic is doomed to talk nonsense – John McCarthy

CHAPTER 1

Counting Is Arithmetic

1.1 Introduction

In this book, we'll learn how to count!

You may be saying: *"But I already know how to count: one, two, three, . . ."*

True. But most counting problems do not involve simply counting a list or group of items. Usually we have to first figure out *what* we're counting, then we have to figure out *how* to count it.

One thing that we will repeat over and over in this chapter, and indeed in the course of the entire book, is:

Important: Don't memorize!



We're going to discuss a lot of counting techniques, and many of them will involve formulas, definitions, or "tricks" that we'll use to solve problems. Don't view this book as a series of problems in which you learn the "trick" for each problem type. Instead, you should learn and understand that all of these counting problems call for appropriate use of addition, subtraction, multiplication, and division.

If you understand what you are adding, subtracting, multiplying, or dividing, and when to do what, then you won't need to memorize 45 different "tricks" for 45 different problems. Instead, you'll know how to do hundreds of different problems because you will *understand* how to count.

We may have some fancy names for some of the techniques that we learn, but remember: at heart, it's just arithmetic. Nothing fancy.

1.2 Counting Lists of Numbers

Problems

Problem 1.1: How many numbers are in the list

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18?

Problem 1.2: How many numbers are in the list

7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29?

In words, we could ask this problem as “How many numbers are there between 7 and 29 inclusive?” (Inclusive means that we include the 7 and the 29 in our count.)

Problem 1.3: Given two positive numbers a and b , with $b > a$, find a formula for how many numbers there are between a and b inclusive. (Remember, inclusive means that we include a and b in our count.)

Problem 1.4: How many multiples of 3 are between 62 and 215?

Problem 1.5:

- (a) How many multiples of 10 are between 9 and 101?
- (b) How many multiples of 10 are between 11 and 103?
- (c) We know that $(101 - 9) = (103 - 11) = 92$, so shouldn't your answers to (a) and (b) be the same? Why aren't they?

Problem 1.6: How many 4-digit numbers are perfect cubes?

We'll start with what is perhaps the simplest counting task: counting lists of numbers. Some lists of numbers are really easy to count.

Problem 1.1: How many numbers are in the list

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18?

Solution for Problem 1.1: Obviously, there are 18 numbers. $\square$

That was pretty easy. The counting was already done for us!

Many other counting problems can be reduced to this type of counting.

Extra! *If people do not believe that mathematics is simple, it is only because they do not realize how complicated life is.* – John von Neumann

Problem 1.2: How many numbers are in the list

7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29?

In words, we could ask this problem as “How many numbers are there between 7 and 29 inclusive?” (**Inclusive** means that we include the 7 and the 29 in our count.)

Solution for Problem 1.2: We could just count them from left to right and find that there are 23 numbers. However, a more clever way to approach this problem is to convert this problem into a problem like Problem 1.1, by subtracting 6 from every number in our list:

$$\begin{array}{cccccc} 7 & 8 & 9 & \cdots & 29 \\ -6 & -6 & -6 & \cdots & -6 \\ \hline 1 & 2 & 3 & \cdots & 23 \end{array}$$

We know how to count the new list! There are 23 items in the list. So, there are 23 items in our original list. $\square$

This problem illustrates a very important problem solving idea.

Concept: When presented with a complicated problem, try to reduce or simplify it to a simpler problem that you know how to solve.



You may also notice that in Problem 1.2, we found that there are $29 - 7 + 1 = 23$ numbers from 7 to 29, inclusive. Perhaps such a formula holds for any two numbers.

Problem 1.3: Given two positive numbers a and b , with $b > a$, find a formula for how many numbers there are between a and b inclusive.

Solution for Problem 1.3: We can subtract $a - 1$ from our list of numbers from a to b to get a list of numbers starting at 1:

$$\begin{array}{cccccc} a & a+1 & a+2 & \cdots & b \\ -(a-1) & -(a-1) & -(a-1) & \cdots & -(a-1) \\ \hline 1 & 2 & 3 & \cdots & b-a+1 \end{array}$$

Our new list then has $b - a + 1$ numbers in it, so our old list does too. So the answer is $b - a + 1$. $\square$

Many other counting problems can be solved using the concept “convert the problem into a simpler problem that we know how to solve.”

Problem 1.4: How many multiples of 3 are between 62 and 215?

Solution for Problem 1.4: We see that $\frac{62}{3} = 20\frac{2}{3}$, so the smallest multiple of 3 in our list is $3 \times 21 = 63$. Similarly, $\frac{215}{3} = 71\frac{2}{3}$, so the largest multiple of 3 in our list is $3 \times 71 = 213$. So our list is

63, 66, 69, ..., 213.

To convert it into a list that we know how to count, we can divide our list by 3:

21, 22, 23, ..., 71.

We know how to count this list! Subtracting 20 from each number in the list gives

$$1, 2, 3, \dots, 51.$$

So there are 51 numbers in the list. $\square$

You might have been tempted to use a little shortcut for Problem 1.4:

Bogus Solution: We can just compute



$$\frac{215 - 62}{3} = \frac{153}{3} = 51$$

to see there are 51 numbers in the list.

But that “shortcut” doesn’t always work very well, as we can see in the next problem:

Problem 1.5:

- (a) How many multiples of 10 are between 9 and 101?
- (b) How many multiples of 10 are between 11 and 103?
- (c) We know that $101 - 9 = 103 - 11 = 92$, so shouldn’t your answers to (a) and (b) be the same? Why aren’t they?

Solution for Problem 1.5: For this problem, it’s easy enough to just list the multiples of 10.

- (a) Our list is

$$10, 20, 30, \dots, 100,$$

so there are 10 multiples.

- (b) Our list is

$$20, 30, \dots, 100,$$

so there are 9 multiples.

- (c) The reason these answers are different is because the “shortcut” solution doesn’t work! You can’t count the number of multiples of 10 simply by calculating

$$\frac{101 - 9}{10} = \frac{103 - 11}{10} = \frac{92}{10} = 9.2.$$

How would you know whether the answer is 9 or 10?

$\square$

WARNING!! Beware of quick shortcuts! (Unless you are absolutely certain that your “shortcut” works.)



Here’s a slightly more unusual list that we can count:

Problem 1.6: How many 4-digit numbers are perfect cubes?

Solution for Problem 1.6: Clearly we don't want to try to solve this problem by testing each 4-digit number to see if it's a perfect cube – that would take forever! Instead, if we can find the smallest and biggest numbers in our list, maybe we can find a convenient way to count all the numbers in between.

The smallest 4-digit perfect cube is $1000 = 10^3$.

The largest 4-digit perfect cube is a little harder to find. This requires a bit of experimentation. We can start by noting that $20^4 = 8000$, and then we can use trial and error:

$$21^3 = 9261$$

$$22^3 = 10648$$

So $9261 = 21^3$ is the largest 4-digit cube. Therefore our list is

$$1000, \dots, 9261.$$

But there's a much better way that we can write this list. Write it as

$$10^3, 11^3, 12^3, \dots, 20^3, 21^3.$$

Clearly the number of numbers in the list above is the same as the number of numbers in the list below:

$$10, 11, 12, \dots, 20, 21.$$

There are 12 numbers in this list, so 12 is our answer. $\square$

Exercises

1.2.1 How many numbers are in the list

$$36, 37, 38, \dots, 92, 93?$$

1.2.2 How many numbers are in the list

$$4, 6, 8, \dots, 128, 130?$$

1.2.3 How many numbers are in the list

$$-33, -28, -23, \dots, 52, 57?$$

Hints: 181

1.2.4 How many numbers are in the list

$$147, 144, 141, \dots, 42, 39?$$

Hints: 118

1.2.5 How many numbers are in the list

$$3\frac{2}{3}, 4\frac{1}{3}, 5, 5\frac{2}{3}, \dots, 26\frac{1}{3}, 27?$$

Hints: 162

1.2.6 How many positive multiples of 7 are less than 150?

1.2.7 How many perfect squares are between 50 and 250?

1.2.8 How many odd perfect squares are between 5 and 211?

1.2.9★ How many sets of four consecutive positive integers are there such that the product of the four integers is less than 100,000? **Hints:** 123, 104

1.3 Counting with Addition and Subtraction

Problems

Problem 1.7: At Brown High School, there are 12 players on the basketball team. All of the players are taking at least one foreign language class. The school offers only Spanish and French as its foreign language classes. 8 of the players are taking Spanish and 5 of the players are taking both languages. How many players are taking French?

Problem 1.8: There are 27 cats at the pound. 14 of them are short-haired. 11 of them are kittens. 5 of them are long-haired adult cats (not kittens). How many of them are short-haired kittens?

Often when we want to count, it's not as simple as creating a list. Many counting problems require us to use a little more thought. Let's look at an example:

Problem 1.7: At Brown High School, there are 12 players on the basketball team. All of the players are taking at least one foreign language class. The school offers only Spanish and French as its foreign language classes. 8 of the players are taking Spanish and 5 of the players are taking both languages. How many players are taking French?

Before we work through the solution (for this problem or for any counting problem), always remember the following:

Important: Don't just blindly add and subtract – think about what you're doing!



Solution for Problem 1.7: The players that are taking French fall into two categories: those who are also taking Spanish, and those who aren't.

The number of players taking French that are also taking Spanish is 5 (this is given in the problem statement).

Next we count the number of players that are taking French but not Spanish. There are 12 players on the team in total, and 8 of them are taking Spanish, so there are $12 - 8 = 4$ members not taking Spanish. Since every player must be taking at least one language, these 4 players are taking French.

So the number of players taking French is the sum of the number of players in each of the two categories, namely $5 + 4 = 9$.

So the answer is 9. $\square$

We can also use a picture to solve this problem:

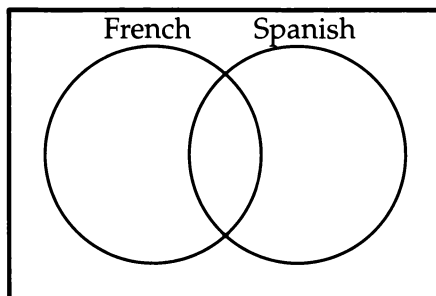


Figure 1.1: Venn diagram

The picture in Figure 1.1 is called a **Venn diagram**. In the diagram we draw a circle for the players taking French and another circle for the players taking Spanish. The circles overlap because some players are taking both languages.

Concept: We can use a Venn Diagram whenever we wish to count things or people which occur in two or three overlapping groups.



We place points in the circles to represent the players – one point per player. A point that is in the French circle that is not in the Spanish one represents a player taking French but not Spanish. A point in the region that is in both circles represents a player taking both languages. A player taking Spanish but not French is represented by a point inside the Spanish circle but not in the French one. Finally, a point placed outside both circles represents a player who is in neither class.

Extra!

John Venn (1834–1923)

➡➡➡➡ The namesake of the Venn diagram was not only a Fellow of Gonville and Caius College of Cambridge University in England, he was also an ordained priest. Venn first published his famous diagram in 1881. In addition to his extensive study of counting and probability, Venn also studied logic and philosophy, and later became an acclaimed historian. Venn was also an accomplished machine builder, and once built a machine
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Now we can use this diagram to solve the problem.

We place five points in the intersection space inside of both circles, because there are five players in both classes. So we start with the diagram in Figure 1.2.

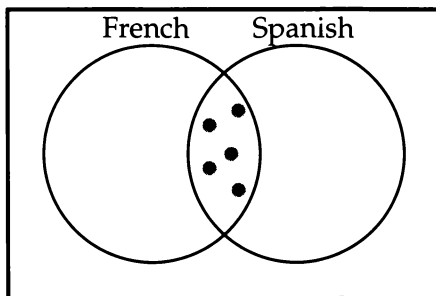


Figure 1.2: Players taking both languages

Now, since there are 8 players taking Spanish, and 5 points are already inside the Spanish circle on the right, there must be three more points inside the Spanish circle which aren't in the French circle. We add these points in Figure 1.3.

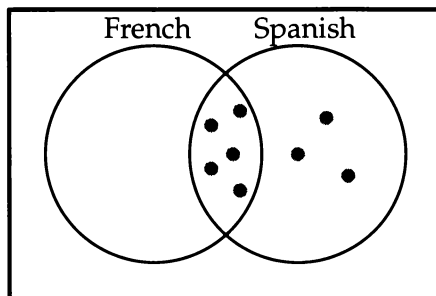


Figure 1.3: Players taking Spanish

Since we have 12 total points and we know there aren't any outside both circles (since there are no players who are not taking either language), there must be 4 left which are inside the French circle but not inside the Spanish circle. After adding these points, we have Figure 1.4.

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➡➡➡➡➡ to bowl cricket balls (which was so well-designed that it outbowled one of the top players on the Australian national cricket team). His contribution to mathematics is memorialized at Cambridge University by a stained-glass window depicting a Venn Diagram – you can view it on our links page at

<http://www.artofproblemsolving.com/BookLinks/IntroCounting/links.php>
 (Source: MacTutor History of Mathematics Archive)

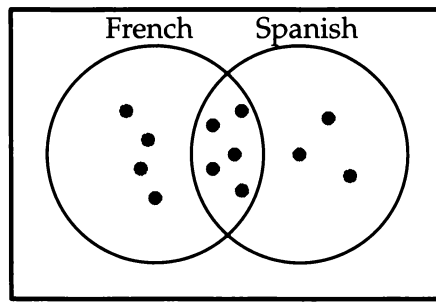


Figure 1.4: All players

We've placed all of the students on the team into the diagram. So now we can just read off the answer to the problem from the diagram in Figure 1.4. There are 9 points inside the French circle on the left, so there are 9 players in the French class.

Obviously, if the numbers in the problem were bigger, it would be a chore to draw all those dots, so usually we use numbers to represent how many points are in each region, as in Figure 1.5.

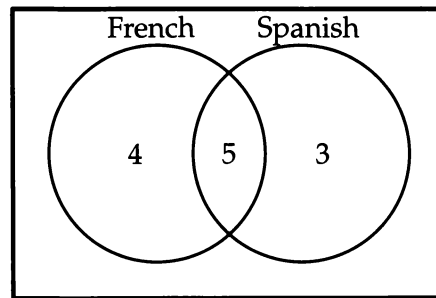


Figure 1.5: All players

Quick question: Why did we start with the 5 players in both classes rather than with the "12 total players" or the "8 players in Spanish"?

We started with the players in both classes because those players are in a single region in our picture – if we started with "8 players in Spanish", we wouldn't know how many to put in the intersection of the two circles and how many to put in the "just Spanish" section.

Important: When using a Venn Diagram, start filling in the diagram with numbers that we know go in a single region of the diagram. Usually, this means that we start in the middle of the diagram.

Let's look at another problem that can be solved using a Venn diagram:

Problem 1.8: There are 27 cats at the pound. 14 of them are short-haired. 11 of them are kittens. 5 of them are long-haired adult cats (not kittens). How many of them are short-haired kittens?

Solution for Problem 1.8: We draw a Venn diagram, with one circle for cats with short hair and one circle for cats which are kittens.

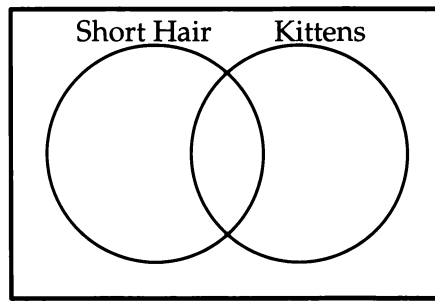


Figure 1.6: Blank Venn diagram for the cats problem

Which number(s) do we want to place in the regions?

Since 5 cats don't have short hair and are not kittens, we know there are 5 cats outside both circles. So we add that to our diagram, giving Figure 1.7.

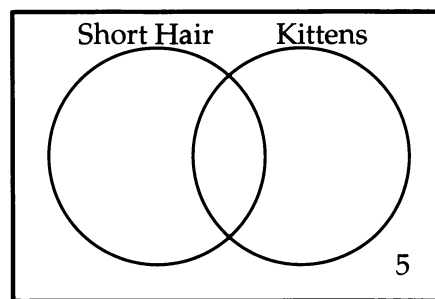


Figure 1.7: Venn diagram for the cats problem

At this point, we can't immediately fill in any of the other numbers, because none of our numbers corresponds exactly to a region of the diagram. For example, we know there are 11 kittens, but there's no single region of the diagram that corresponds to "kittens": there's a region for "short-haired kittens" and a region for "long-haired kittens." So we're going to have to use a little bit of thought.

We'll have to introduce a variable. We can call the number of cats in one of the regions inside the circles x , and try to find other regions in terms of x . We do so because we do have information about combinations of regions of the circles; we just don't know any one specific region.

Let the number of short-haired kittens (which are in the intersection of the circles) be x . Note that we chose x to be the answer to the problem.

Concept: When assigning a variable, it's usually best to let your variable represent the answer to the problem.



We place x into our Venn diagram as in Figure 1.8.

Extra! *Each problem that I solved became a rule, which served afterwards to solve other problems.*

➡➡➡➡ – Rene Descartes

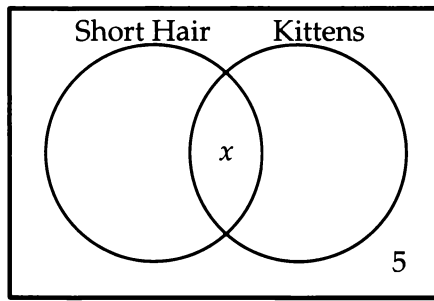


Figure 1.8: Venn diagram with a variable entry

Can we place expressions in the other two regions?

Since there are a total of 14 short-haired cats, and x of them are kittens, we know that $14 - x$ of them are not kittens. Therefore, we place $14 - x$ in the portion of the short-haired circle that does not overlap with the kittens. Similarly, we have $11 - x$ kittens which are not short-haired. So our Venn diagram now has an entry in every region, as shown in Figure 1.9.

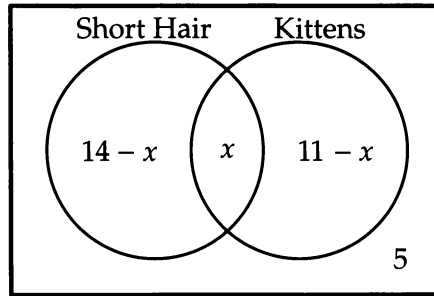


Figure 1.9: Venn diagram with all entries filled

There's one more piece of information that we haven't used yet, and that's the total number of cats, which is 27. So if we add everything in our diagram, we must get 27:

$$(14 - x) + (11 - x) + x + 5 = 27.$$

This simplifies to $30 - x = 27$, so $x = 3$ is our answer. $\square$

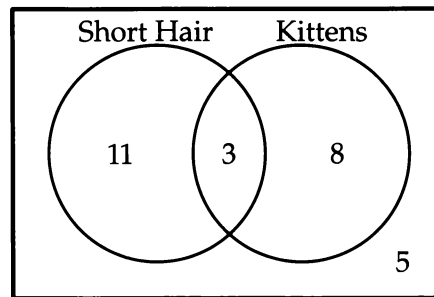


Figure 1.10: Venn diagram with all the numbers

If we wanted to fill up our Venn diagram from Figure 1.9 with numbers, it would look like Figure 1.10.

One advantage of using this diagram is that not only can we read our answer directly from the diagram (there are 3 short-haired kittens), but we can also easily check that our solution matches the problem description, and therefore know that we didn't make a mistake. At a glance, we can confirm that there are 14 short-haired cats, 11 kittens, and 27 cats total.

**Concept:** If it's easy to do, it's always a good idea to check that your answer is consistent with the problem statement.

Of course, we could also do this problem without a diagram. You will eventually (and may already) do problems of this nature without a diagram; however, you'll likely at least visualize a diagram or a table, which will help keep the problem clear and prevent you from making careless errors.

Exercises

1.3.1 There are 20 cars in my building's parking lot. All of the cars are red or white. 12 of them are red, 15 of them are 4-door, and 4 of them are 2-door and white. How many of the cars are 4-door and red?

1.3.2 Going back to the 12-person basketball team from Problem 1.7, all 12 players are taking at least one of biology or chemistry. If 7 players are taking biology and 2 players are taking both sciences, how many players are taking chemistry?

1.3.3 There are 30 students in Mrs. Taylor's kindergarten class. If there are twice as many students with blond hair as with blue eyes, 6 students with blond hair and blue eyes, and 3 students with neither blond hair nor blue eyes, how many students have blue eyes? **Hints:** 175

1.3.4★ At the Gooddog Obedience School, dogs can learn to do three tricks: sit, stay, and roll over. Of the dogs at the school:

50 dogs can sit	17 dogs can sit and stay
29 dogs can stay	12 dogs can stay and roll over
34 dogs can roll over	18 dogs can sit and roll over
9 dogs can do all three	9 dogs can do none

- (a) How many dogs are in the school? **Hints:** 141
- (b) How many dogs can do exactly 2 tricks? **Hints:** 131

1.3.5★ Every student in my school is in either French class or Spanish class, or both. Let x be the number of students in French class, y be the number of students in Spanish class, and z be the number of students that are in both classes. Find an expression in terms of x , y , and z for how many students there are in my school. **Hints:** 157

1.4 Counting Multiple Events

Problems

Problem 1.9: You have three shirts and four pairs of pants. How many outfits consisting of one shirt and one pair of pants can you make?

Problem 1.10: In how many ways can we form an international commission if we must choose one European country from among 6 European countries, one Asian country from among 4, one North American country from among 3, and one African country from among 7?

Problem 1.11: In how many ways can we form a license plate if there are 7 characters, none of which is the letter O, the first of which is a numerical digit (0-9), the second of which is a letter, and the remaining five of which can be either a digit or a letter (but not the letter O)?

Problem 1.12: In how many ways can I arrange four different books on a shelf?

Problem 1.13: Your math club has 20 members. In how many ways can it select a president, a vice-president, and a treasurer if no member can hold more than one office?

Problem 1.14: In how many ways can we arrange n different books, where n is a positive integer?

Often we'll be counting the number of outcomes of a series of events. Here's an example of this type of problem.

Problem 1.9: You have three shirts and four pairs of pants. How many outfits consisting of one shirt and one pair of pants can you make?

Solution for Problem 1.9: In this problem, the number of possibilities is so small that we can just list them outright.

If our shirts are labeled S_1, S_2, S_3 and our pants are labeled P_1, P_2, P_3, P_4 , then we can list all of the possible outfits:

$$S_1P_1, S_1P_2, S_1P_3, S_1P_4, S_2P_1, S_2P_2, S_2P_3, S_2P_4, S_3P_1, S_3P_2, S_3P_3, S_3P_4$$

So there are 12 outfits. $\square$

Listing the outfits like this is somewhat annoying. We can better visualize the outfits by drawing a tree as in Figure 1.11, or a grid as in Figure 1.12.

Extra! *There is no branch of mathematics, however abstract, which may not some day be applied to phenomena of the real world.* – Nikolai Lobachevsky

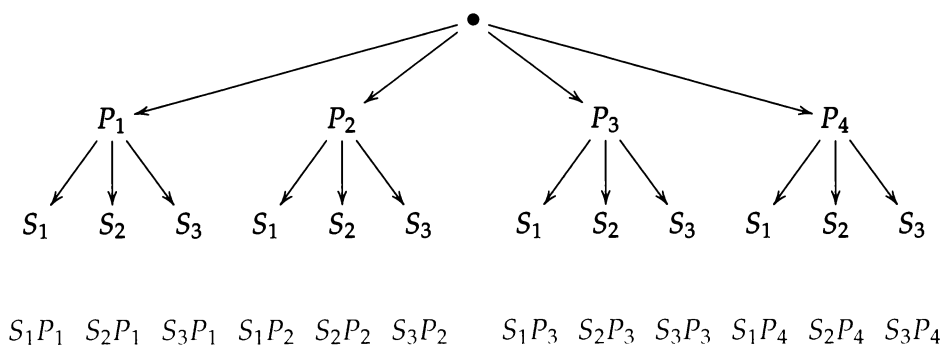


Figure 1.11: Outfits as a tree

		Pants			
		P_1	P_2	P_3	P_4
Shirts	S_1	S_1P_1	S_1P_2	S_1P_3	S_1P_4
	S_2	S_2P_1	S_2P_2	S_2P_3	S_2P_4
	S_3	S_3P_1	S_3P_2	S_3P_3	S_3P_4

Figure 1.12: Outfits as a grid

For both of these pictures, we can use the following reasoning: we have four choices for the pants, and for each of these four choices of pants, we have three choices for the shirt. Therefore, there are $4 \times 3 = 12$ outfits.

Concept: The point of drawing a tree or a grid is to keep our counting organized. Clear organization is very important in solving counting problems.

Drawing a tree or a grid is easy to do in this problem because the numbers are small. Let’s look at a more complicated version of the same type of problem.

Problem 1.10: In how many ways can we form an international commission if we must choose one European country from among 6 European countries, one Asian country from among 4, one North American country from among 3, and one African country from among 7?

Solution for Problem 1.10: Clearly drawing a grid or a tree is not going to be practical – there are too many choices! So we’ll have to do this problem by reasoning out the answer.

We tackle a problem like this in steps.

Step 1: There are 6 ways to choose a European country.

Step 2: For each European country, we can choose an Asian country 4 ways, for a total of $6 \times 4 = 24$ ways to choose both a European and an Asian country.

Step 3: For each pair of countries that we’ve chosen in Steps 1 and 2, we can choose a North American

country in 3 ways. So there are $24 \times 3 = 72$ ways to choose 3 countries.

Step 4: For each triple of countries that we've chosen in Steps 1–3, we can choose an African country in 7 ways. So there are $72 \times 7 = 504$ ways to choose 4 countries.

So we see that there are $6 \times 4 \times 3 \times 7 = 504$ possibilities. $\square$

Concept: We use multiplication to count a series of independent events.



By **independent**, we mean that each decision does not depend on the others. For example, in the previous problem, we choose a European country, then an Asian country, then a North American country, then an African country. Each choice doesn't affect the other choices, so they are **independent**.

Problem 1.11: In how many ways can we form a license plate if there are 7 characters, none of which is the letter O, the first of which is a numerical digit (0–9), the second of which is a letter, and the remaining five of which can be either a digit or a letter (but not the letter O)?

Solution for Problem 1.11: Since each character does not depend on any of the other characters, our choices are independent. There are 10 choices for the first character (any digit from 0 through 9), there are 25 choices for the second character (any letter A–Z except for O), and there are 35 choices for each of the other five characters (any digit 0–9 or any letter A–Z, except O).

Therefore, since the choices are independent, we have

$$10 \times 25 \times 35 \times 35 \times 35 \times 35 \times 35 = 10 \times 25 \times 35^5 = 13,130,468,750$$

ways to form our license plate. $\square$

In some counting problems, we'll be making a series of choices, but later choices will depend on some of the earlier choices. This is unlike the problems that we've done before. For example, in Problem 1.9, our choices of shirt didn't depend at all on our choice of pants. Things are a little bit different in the next problem.

Problem 1.12: In how many ways can I arrange four different books on a shelf?

Solution for Problem 1.12: We cannot just count $4 \times 4 \times 4 \times 4$, because once we place the first book, we only have 3 choices left for the second book. So we'll carefully count our choices step by step:

Step 1. We have 4 choices for the first book.

Step 2. We have 3 books remaining, so we have 3 choices for the second book. Thus we have a total of $4 \times 3 = 12$ choices for the first two books.

Step 3. We have 2 books remaining, so we have 2 choices for the third book. Thus we have a total of $12 \times 2 = 24$ choices for the first three books.

Step 4. We only have 1 book remaining, so we have 1 choice for the fourth book. Thus we have a total of $24 \times 1 = 24$ choices for all four books.

So the answer is 24. $\square$

One important thing to note about Problem 1.12 is that although the choices themselves are not independent at each step, the *number* of choices at each step is independent of our previous choices. For example, no matter which book we choose in Step 1, we always have 3 remaining books to choose from in Step 2, even though the specific books that we're choosing from differ based on our choice in Step 1. This means that we can get our answer by multiplying the number of choices at each step, so that $4 \times 3 \times 2 \times 1 = 24$ is the answer.

We could represent our choices in this problem as a tree, where the books are labeled A, B, C , and D , as shown in Figure 1.13 on the next page.

Here's another example of a counting problem in which the choices are not independent.

Problem 1.13: Your math club has 20 members. In how many ways can it select a president, a vice-president, and a treasurer if no member can hold more than one office?

Solution for Problem 1.13: Once again, our choices are not independent. Once a student has been chosen president, she is not available to be chosen vice-president or treasurer. However, the number of choices we have for each position is the same no matter who is chosen.

We have 20 choices for president. Once we've chosen a president, then we have 19 people remaining to choose from for vice-president. After we've chosen both president and VP, we then have 18 people remaining to choose from for treasurer.

Therefore, there are $20 \times 19 \times 18 = 6840$ ways to fill the three offices. $\square$

The last two problems are examples of what we call **permutations**. A permutation occurs whenever we have to choose several items one at a time from a larger group of items.

In Problem 1.12, we are asked for the number of permutations of four books – in other words, how many ways we could order the four different books.

In Problem 1.13, we are asked for the number of permutations of 3 people out of 20 people, meaning we want to count the number of ways to fill three different slots with people from a group of 20 people. We also can say that this is the number of permutations of size 3 of the group of 20 people.

Important: Don't start thinking of these problems as:



"problem" $\rightarrow$ "permutation" $\rightarrow$ "multiply"

You should be able to skip that vocabulary step in the middle. Vocabulary crutches lead to memorization. Instead, understand why you multiply; don't multiply because "that's what you do on permutation problems." Instead of "problem" $\rightarrow$ "permutation" $\rightarrow$ "multiply", you should just think "problem" $\rightarrow$ "multiply" when you see a permutation problem.

In Problem 1.12, we saw that the number of permutations of 4 books is $4 \times 3 \times 2 \times 1$. Let's look at the general version of this problem.

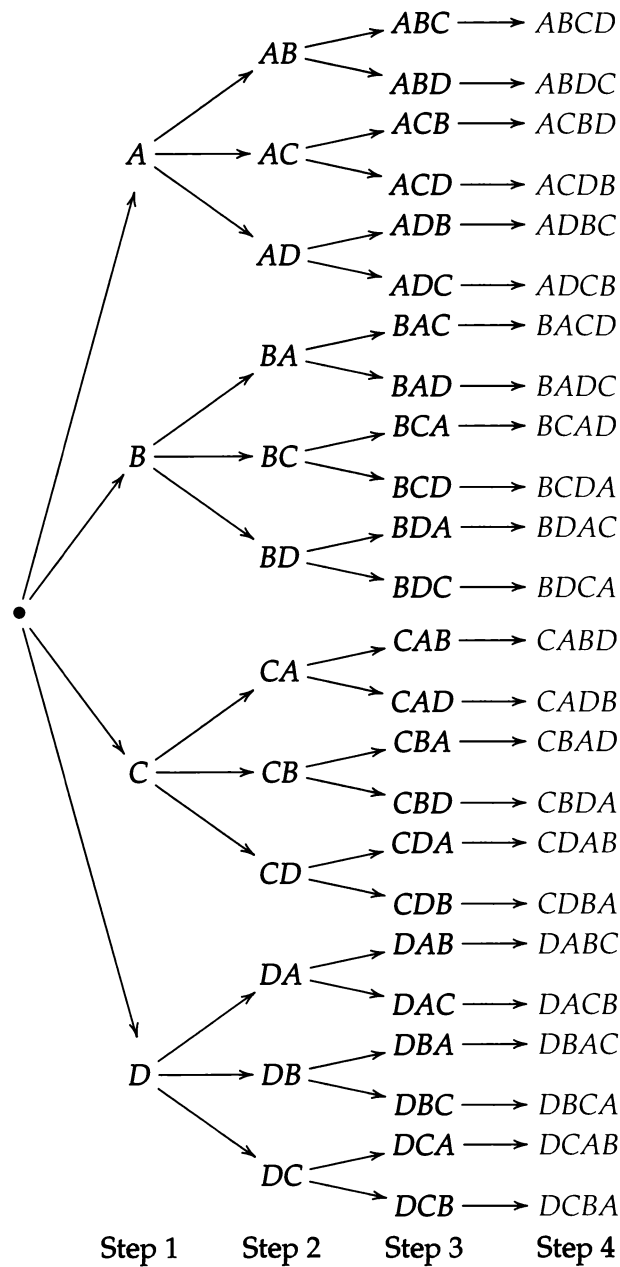


Figure 1.13: Tree for choosing 4 books

Problem 1.14: In how many ways can we arrange n different books, where n is a positive integer?

We call this “solving the general problem,” since if we find an expression for the answer in terms of n , we can solve any specific problem just by plugging n into our “general formula.”

WARNING!!  Don't blindly memorize formulas. If you understand them, there's no need to memorize – you'll know instead.

Solution for Problem 1.14: As we did before, we approach this problem step by step.

There are n choices for the first book.

Once we place the first book, there are $n - 1$ books remaining, so there are $n - 1$ choices for the second book.

Once we place the second book, there are $n - 2$ books remaining, so there are $n - 2$ choices for the third book.

And so on, until we are eventually left with just 1 choice for the n^{th} and final book.

So there are

$$n \times (n - 1) \times (n - 2) \times (n - 3) \times \cdots \times 3 \times 2 \times 1$$

choices for ordering all n books. $\square$

The quantity that is the answer to Problem 1.14 occurs so frequently in these types of problems (and indeed in mathematics in general) that it has a special name and notation. We call this quantity **n factorial** and denote it by $n!$. So $n!$ is equal to the product of the first n positive integers.

Important: $n! = n \times (n - 1) \times (n - 2) \times (n - 3) \times \cdots \times 3 \times 2 \times 1$.



For example,

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120,$$

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720,$$

and so on.

In summary, we can order n different objects in $n!$ ways. But as we said before, don't memorize this! If you understand it, you'll never have to memorize – it will come automatically. If you have to memorize it, then you don't understand it and should think about it some more.

One more thing about permutations: it's convenient to have a definition for $0!$, which is the number of ways to arrange 0 objects in a row. But there's only one way to arrange zero objects in a row: do nothing! So $0! = 1$.

Exercises

1.4.1 Suppose I have 6 shirts and 5 ties. How many shirt-and-tie outfits can I make?

1.4.2 For each of 8 colors, I have one shirt and one tie of that color. How many shirt-and-tie outfits can I make if I refuse to wear a shirt and a tie of the same color?

1.4.3 How many license plates consist of 2 letters followed by 3 numbers?

1.4.4 How many license plates consist of 3 letters, followed by 2 even digits, followed by 2 odd digits?

1.4.5 In how many ways can I stack 5 books on a shelf?

1.4.6 Suppose that I have 6 different books, 2 of which are math books. In how many ways can I stack my 6 books on a shelf if I want a math book on both ends of the stack? **Hints:** 1

1.4.7 There are 8 sprinters in the Olympic 100-meter finals. The gold medal goes to first place, silver to second, and bronze to third. In how many ways can the medals be awarded?

1.4.8 My club has 15 members. In how many ways can we choose a president, vice-president, secretary, and treasurer, if no member can hold more than one office?

1.4.9 Compute each of the following without using a calculator:

- (a) $9!/8!$
- (b) $42!/40!$
- (c) $8! - 7!$ **Hints:** 201

1.5 Permutations

Problems

Problem 1.15: A club has n members, where n is a positive integer. We want to choose r different officers of the club (where r is also a positive integer, and $r < n$), such that no member can hold more than one office.

- (a) In how many ways can we fill the first office?
- (b) Once we've filled the first office, in how many ways can we fill the second office?
- (c) If we proceed as in (a) and (b) above, in how many ways can we fill the r^{th} office?
- (d) In how many ways can we choose the r officers?

Problem 1.16: My city is running a lottery. In the lottery, 25 balls numbered 1 through 25 are placed in a bin. Four balls are drawn one at a time and their numbers are recorded. The winning combination consists of the four selected numbers in the order they are selected. How many winning combinations are there, if:

- (a) each ball is discarded after it is removed?
- (b) each ball is replaced in the bin after it is removed and before the next ball is drawn?

Let's now try to generalize permutations. We'll try a general version of Problem 1.13:

Problem 1.15: A club has n members, where n is a positive integer. In how many ways can we choose r different officers of the club (where r is also a positive integer, and $r < n$), such that no member can hold more than one office?

Solution for Problem 1.15: As usual, we count our choices step by step.

We have n choices to fill the first office.

We have $n - 1$ choices left to fill the second office.

We have $n - 2$ choices left to fill the third office.

And so on. When does the process stop?

When we get to the r^{th} office, we've already chosen $r - 1$ members for the previous $r - 1$ offices, so we have $n - (r - 1) = n - r + 1$ choices for the r^{th} office.

Therefore, we have

$$n \times (n - 1) \times (n - 2) \times \cdots \times (n - r + 2) \times (n - r + 1)$$

ways to fill the r offices. $\square$

This quantity is called **the number of permutations of size r from a group of n objects**, and is denoted $P(n, r)$:

$$P(n, r) = n \times (n - 1) \times (n - 2) \times \cdots \times (n - r + 2) \times (n - r + 1).$$

Is there an easier way to write this quantity? Notice that it looks a lot like $n!$, except that some of terms are missing. Precisely which terms are missing?

Looking at $P(n, r)$ carefully, we see that it's $n!$ with the $(n - r)!$ terms missing:

$$P(n, r) = \frac{n \times (n - 1) \times (n - 2) \times (n - 3) \times \cdots \times (n - r + 1) \times (n - r) \times (n - r - 1) \times \cdots \times 3 \times 2 \times 1}{(n - r) \times (n - r - 1) \times \cdots \times 3 \times 2 \times 1} = \frac{n!}{(n - r)!}.$$

Again, don't memorize – understand. When faced with counting the number of ways to pick 3 officers from 30 people, you should think “30 for the first, 29 for the second, 28 for the third, so $30 \times 29 \times 28$.” This gives you the answer quickly. If you blindly plug the numbers into the formula

$$P(30, 3) = \frac{30!}{27!},$$

then you're not really understanding the problem. Besides, $30!$ and $27!$ are awfully hard to compute!

Let's look at one more problem that illustrates some of the subtleties in counting multiple events.

Problem 1.16: My city is running a lottery. In the lottery, 25 balls numbered 1 through 25 are placed in a bin. Four balls are drawn one at a time and their numbers are recorded. The winning combination consists of the four selected numbers in the order they are selected. How many winning combinations are there, if:

- (a) each ball is discarded after it is removed?
- (b) each ball is replaced in the bin after it is removed and before the next ball is drawn?

Solution for Problem 1.16: (a) We have 25 choices for the first ball. After the first ball is drawn and discarded, there are 24 balls left in the bin, so there are 24 choices for the second ball. Similarly, there are 23 choices for the third ball and 22 choices for the fourth ball. Hence the number of winning combinations is $25 \times 24 \times 23 \times 22 = 303,600$.

(b) We have 25 choices for each of the four balls, since when each ball is drawn, all 25 balls are in the bin. Hence there are $25 \times 25 \times 25 \times 25 = 25^4 = 390,625$ winning combinations. $\square$

The difference between these two examples is the difference between making selections *without* replacement and *with* replacement. In part (a), we don't place the balls back in the bin – we end up with a permutation problem. In part (b), we place each ball back in the bin after it is selected, and therefore have more choices for each number.

Once again, there's nothing to memorize here – don't memorize that “no replacement = permutation.” Just be aware when you are looking at a problem that there's a difference between doing something with replacement and without. As we've just seen, both types of problem call for relatively straightforward multiplication.

Exercises

1.5.1 Compute the following:

- (a) $P(8, 3)$
- (b) $P(20, 4)$
- (c) $P(30, 1)$
- (d) $P(6, 5)$
- (e) $P(50, 3)$

1.5.2 What is $P(n, n)$ for any positive integer n ? **Hints:** 191

1.5.3 12 balls numbered 1 through 12 are placed in a bin. In how many ways can 3 balls be drawn, in order, from the bin, if:

- (a) each ball remains outside the bin after it is drawn?
- (b) each ball is placed back into the bin immediately after it is drawn?
- (c) the first ball is replaced after it is drawn but the second ball remains outside the bin?

1.5.4★ Can you simplify $P(n, k)P(n - k, j)$ where n, k and j are integers such that $k \leq n$ and $j \leq n - k$? **Hints:** 94

1.6 Summary

- Counting a list of numbers like 1, 2, 3, ..., 38 is easy! Counting most other lists of numbers is easy too: use arithmetic to convert it into a list that looks like 1, 2, 3, But beware of “shortcuts” to counting such lists that don't actually work!

- The number of items in the list $a, a + 1, \dots, b - 1, b$, where $a \leq b$ are integers, is $b - a + 1$. But don't memorize this formula – understand how it is derived.
- Counting items which lie in one or more overlapping sets requires a thoughtful use of addition and subtraction. We can use a Venn Diagram to help keep track of the thought process. When using a Venn Diagram, we should initially fill in the numbers which correspond exactly to a single region of the diagram. We may have to place variables in our diagram: if so, it's usually wise to choose our variable to represent the answer to the problem.
- When counting multiple events, we multiply: the number of ways that A **and** B can occur is
 (the number of ways A can occur) **times** (the number of ways B can occur).

Sometimes our later choices are not independent of our earlier choices, so our counting must take that into consideration.

- We can arrange n items in $n!$ ways, where

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 3 \times 2 \times 1.$$

Again, don't simply memorize this formula; understand why it works!

- We can select r items in order from a group of n items in $P(n, r)$ ways, where

$$P(n, r) = n \times (n - 1) \times \cdots \times (n - r + 2) \times (n - r + 1) = \frac{n!}{(n - r)!}.$$

We say again: don't memorize this, understand it!

Some useful ideas:

Important: Don't memorize!



Concept: When presented with a complicated problem, try to reduce or simplify it to a simpler problem that you know how to solve.



WARNING!! Beware of quick shortcuts! (Unless you can prove that your "shortcut" works.)



Important: Don't just blindly add and subtract – think about what you're doing!



Concept: When assigning a variable, it's usually best to let your variable represent the answer to the problem.

Concept: The point of drawing a tree or a grid is to keep our counting organized. Clear organization is very important in solving counting problems.

Concept: If it's easy to do, it's always a good idea to check that your answer is consistent with the problem statement.

REVIEW PROBLEMS



1.17 How many numbers are in the list

2.5, 5.5, 8.5, 11.5, ..., 80.5, 83.5?

1.18 How many numbers are in the list

6, 10, 14, ..., 82, 86?

1.19 How many 3-digit numbers are divisible by 7?

1.20 There are 20 people in my club. 8 of them are left-handed. 15 of them like jazz music. 2 of them are right-handed and dislike jazz music. How many club members are left-handed and like jazz music?

1.21 Mr. Stewart's math class has 16 boys, and 22 of his students bring their calculators to class. If 13 of the students with calculators are girls, then how many boys didn't bring their calculators to class?

1.22 You have 5 shirts, 6 pairs of pants, and 8 hats. How many outfits can you make consisting of one shirt, one pair of pants, and one hat?

1.23 How many 3-letter combinations can be formed if the second letter must be a vowel (A, E, I, O, or U), and the third letter must be different from the first letter?

1.24 The local theater has one ticket window. In how many ways can six people line up to buy a ticket? (Source: MATHCOUNTS)

1.25 In how many ways can you order 7 different colored hats in a row?

1.26 Our basketball team has 12 members, each of whom can play any position. In how many ways can we choose a starting lineup consisting of a center, a power forward, a shooting forward, a point guard, and a shooting guard?

1.27

- (a) List all the arrangements of the letters in the word 'T'. How many are there?
- (b) List all the arrangements of the letters in the word 'TI'. How many start with 'T'? How many with 'I'? How many are there?
- (c) List all the arrangements of the letters in the word 'ETI'. Circle the ones that start with 'E' and compare that group to the list in part (b). How many arrangements of 'ETI' start with 'E'? With 'T'? With 'I'? How many arrangements are there total?
- (d) List all the arrangements of the letters in the word 'SETI'. Circle the ones that start with 'S' and compare that group to the list in part (c). How many arrangements of 'SETI' start with 'S'? How many arrangements are there total?
- (e) How many arrangements are there of the letters in the word 'ASETI'?

1.28 In the upcoming election, voters can choose:

- Abe or Barb for President;
 - Carlos, Darla, or Eve for Vice-President;
 - Fred, Gina, or Hal for Secretary.
- (a) Draw a tree demonstrating the possible ways a voter might complete his or her ballot.
 - (b) How many different ways can the ballot be completed if each voter must cast a vote for each office?
 - (c) How many ways can the ballot be completed if the voter may choose not to vote in some (or all) of the elections?

Challenge Problems



1.29 How many positive integers less than 500 can be written as the sum of two positive perfect cubes?

Hints: 35, 65

1.30 There are 190 people on the beach. 110 are wearing sunglasses, 70 are wearing bathing suits, and 95 are wearing a hat. Everyone is wearing at least one of these items. 30 are wearing both bathing suits and sunglasses. 25 are wearing both bathing suits and a hat. 40 are wearing both sunglasses and a hat. How many people are wearing all three items?

1.31 There are 360 people in my school. 15 take calculus, physics, and chemistry, and 15 don't take any of them. 180 take calculus. Twice as many students take chemistry as take physics. 75 take both calculus and chemistry, and 75 take both physics and chemistry. Only 30 take both physics and calculus. How many students take physics? Hints: 138

1.32★ In my fencing club there are twice as many boys who are in high school as there are boys who are in middle school. Of the students in the club who are in middle school, there are three times as many

girls as there are boys. Half of the girls who are in the club are in high school. If there are 72 people in my fencing club, how many are middle school boys? **Hints:** 49, 4

1.33 What is the greatest common factor of $5!$, $10!$, and $15!$? **Hints:** 24

1.34★ What is the units digit of the sum $1! + 2! + 3! + 4! + 5! + \cdots + 1000!$? **Hints:** 185, 212

1.35 How many of the factorials from $1!$ to $100!$ are divisible by 9? **Hints:** 32

1.36 Which integers n satisfy $\frac{1}{2} > \frac{1}{n} > \frac{3}{100}$, and how many such integers are there?

1.37 My classroom has 11 rows of chairs, with 11 chairs in each row. The chairs in each row are numbered from 1 to 11.

(a) How many chairs have odd numbers?

(b)★ Suppose we replaced 11 with n . Can you find a formula in terms of n for the number of chairs with odd numbers? **Hints:** 219

1.38 We connect dots with toothpicks in a grid as shown in Figure 1.14. If there are 10 horizontal toothpicks in each row and 20 vertical ones in each column, how many total toothpicks are there? **Hints:** 150 (Source: AHSME)

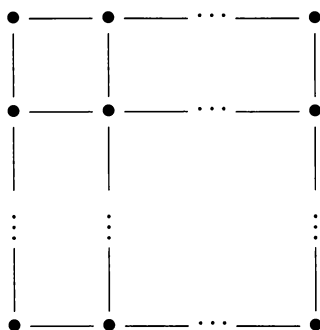


Figure 1.14: Figure for Problem 1.38

